1394 Interface Implementation Guideline for D-VHS

Version 1.00

May 29, 2000

VHS Standard Center

JVC

Contents

1 Purpose

The purpose of this document is to guarantee the basic inter-operability of D-VHS, which implements 1394 digital-interface.

2 References

The following normative documents contain provisions, which, through reference in this document, constitute provisions of this guideline.

[R1] IEEE Std 1394-1995, Standard for a High Performance Serial Bus
[R2] IEEE Std 1394a-2000, IEEE Standard for a High Performance Serial Bus--Amendment 1
[R3] IEC 61883-1, Consumer audio/video equipment - Digital interface- Part 1: General
[R4] IEC 61883-4, Consumer audio/video equipment – Digital interface – Part 4: MPEG2-TS data transmission
[R5] 1394TA, Configuration ROM for AV/C Devices 1.0
[R6] 1394TA, AV/C Digital Interface Command Set, General Specification Version 3.0
[R7] 1394TA, AV/C Digital Interface Command Set, Tape Recorder/Player Subunit Specification Version 2.1
[R8] Microsoft, Plug and Play Design Specification for IEEE1394 Version 1.0c

3 Clarifications

3.1 High performance serial bus layers

3.1.1 Cable physical layer

All cable physical layer implementations conforming to this standard shall meet the performance criteria specified in IEC 61883-1, Clause 4.1 [R3].

3.1.1.1 Connector and cable

4pin AV cable is recommended for connecting D-VHS.

3.1.1.2 Data rate capability

Data rate capability is recommended to support S200 or more.

NOTE: If D-VHS can know data rate capabilities of all nodes connected the bus, D-VHS may use the lowest data rate to establish broadcast-out connection.

3.1.2 Link layer

All link layer implementations conforming to this guideline shall meet the performance criteria specified in IEC 61883-1, Clause 4.2 [R3].

3.1.3 Transaction layer

All transaction layer implementations conforming to this guideline shall meet the performance criteria specified in IEC 61883-1, Clause 4.3 [R3].

3.2 Serial bus management

All implementations regarding basic serial bus management conforming to this guideline shall meet the performance criteria specified in IEC 61883-1, Clause 5 [R3].
Implementation requirements for the configuration ROM format in this guideline conform to IEC61883-1, Clause 5.3.3 [R3] and IEEE Std 1394a-2000, Clause 10.25 and 10.26 [R2].

In addition, the following items shall be added.

- Vendor Name
- Model ID
- Model Name

Implementation for those three items above conforms to the following document.

- 1394 Trade Association, Configuration ROM for AV/C Devices 1.0 [R5].

A string length of Vendor Name and Model Name should be within 16 bytes.
The example of D-VHS configuration ROM format is shown in Clause 3.10.

NOTE: The configuration ROM format of this guideline would be updated as that defined in the 1394 Trade Association.

3.3 Real time data transmission protocol

Real time data transmission protocol implementations conforming to this guideline shall meet the performance criteria specified in IEC 61883-1, Clause 6 [R3].
If D-VHS shall transmit an isochronous packet but no data to transmit is available, an empty packet shall be transmitted instead of a no data packet (FMT=111111_2).

3.4 Isochronous data flow management

Isochronous data flow management implementations conforming to this guideline shall meet the performance criteria specified in IEC 61883-1, Clause 7 [R3].

3.5 Connection management procedures (CMP)

Connection management procedures implementations conforming to this guideline shall meet the performance criteria specified in IEC 61883-1, Clause 8 [R3].
In addition to CMP, the following rules are defined in this guideline.

D-VHS shall implement functionality that

1) can select an isochronous transmitter (source) from isochronous capable node(s) on the bus regardless whether that Source node outputs isochronous data or not and
2) can establish a point-to-point connection between an oPCR of the selected Source node and its own iPCR.

NOTE: If D-VHS de-selects the Source node, it should break the point-to-point connection. When D-VHS is in power-off mode, no source selection should be left.

3.6 Function Control Protocol (FCP)

Function Control Protocol implementations conforming to this guideline shall meet the performance criteria specified in IEC 61883-1, Clause 9 [R3].
D-VHS implements AV/C Digital Interface Command Set defined in 1394 Trade Association. The following two AV/C specifications shall be supported.

1. General Specification Version 3.0 [R6]
2. Tape Recorder/Player Subunit Specification Version 2.1 [R7]

In the 1394TA specifications, three kinds of command support levels, Mandatory, Recommended and Optional are defined.
Based on those two specifications, this guideline classifies these three support levels into only two levels, Mandatory and Optional, so that D-VHS interoperability can be easily reached.

3.7 AV/C Command Set

3.7.1 D-VHS Support Command

The mandatory level command shall be executed if D-VHS is in a condition where that command can be executed.
In the following table, an asterisk in the control command column indicates that the command operands determine whether the command is D-VHS Mandatory (M), or D-VHS Optional (O).

M: D-VHS Mandatory O: D-VHS Optional --: Not Defined
Optional command area (includes NOT Defined Area)

Unit Command	Value	Control	Status	Notify
CHANNEL USAGE	12h	--	O	O
CONNECT	24h	*	M	O
CONNECT AV	20h	O	O	O
CONNECTIONS	22h	--	O	--
DIGITAL INPUT	11h	O	O	--
DIGITAL OUTPUT	10h	O	O	--
DISCONNECT	25h	O	--	--
DISCONNECT AV	21h	O	--	--
INPUT PLUG SIGNAL FORMAT	19h	M	M	O
OUTPUT PLUG SIGNAL FORMAT	18h	M	M	O
SUBUNIT INFO	31h	--	M	--
UNIT INFO	30h	--	M	--
OPEN DESCRIPTOR	08h	O	O	O
READ DESCRIPTOR	09h	O	--	--
WRITE DESCRIPTOR	0Ah	O	O	--
SEARCH DESCRIPTOR	0Bh	O	--	--
OBJECT NUMBER SELECT	0Dh	O	O	O
POWER	B2h	*	M	M
RESERVE	01h	O	O	O
PLUG INFO	02h	--	O	--

Table 3-1 Unit commands

-

Tape Recorder/Player Subunit command	Value	Control	Status	Notify
ANALOG AUDIO OUTPUT MODE	70h	O	O	--
AREA MODE	72h	O	O	--
ABSOLUTE TRACK NUMBER	52h	O	O	--
AUDIO MODE	71h	O	O	--
BACKWARD	56h	O	--	--
BINARY GROUP	5Ah	--	--	--
EDIT MODE	40h	O	O	--
FORWARD	55h	O	--	--
INPUT SIGNAL MODE	79h	M	M	--
LOAD MEDIUM	C1h	O	--	--
MARKER	CAh	O	O	O
MEDIUM INFO	DAh	--	M	--
OPEN MIC	60h	--	--	--
OUTPUT SIGNAL MODE	78h	M	M	--
POWER	B2h	O	O	O
PLAY	C3h	*	--	--
PRESET	45h	O	O	--
READ MIC	61h	--	--	--
RECORD	C2h	*	--	--
RECORDING DATE	53h	O	O	--
RECORDING SPEED	DBh	M	M	--
RECORDING TIME	54h	--	O	--
RELATIVE TIME COUNTER	57h	M	M	--
SEARCH MODE	50h	--	O	O
SMPTE/EBU RECORDING TIME	5Ch	--	--	--
SMPTE/EBU TIME CODE	59h	--	--	--
TAPE PLAYBACK FORMAT	D3h	O	M	--
TAPE RECORDING FORMAT	D2h	M	M	--
TIME CODE	51h	O	O	--
TRANSPORT STATE	D0h	--	M	M
WIND	C4h	M	--	--
WRITE MIC	62h	--	--	--

Table 3-2 Tape recorder/player subunit commands

VENDOR-DEPENDENT Command (Defined as Unit Command)	Value	Control	Status	Notify
TIMER	00h	--	O	--
VERSION	00h	--	M	--

Table 3-3 VENDOR-DEPENDENT Unit commands

3.7.1.1 Clarification of D-VHS Support Commands

Support levels: M: Mandatory R: Recommended O: Optional --: Not Defined

The mandatory level command shall be executed if D-VHS is in a condition where that command can be executed.

Unit command	opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
Control command				
CONNECT	24h	O	M	a) If D-VHS has recording capability then the CONNECT control command whose destination_subunit_type field specifies a tape recorder/player subunit is mandatory b) If D-VHS in a playback mode can output a signal from other than the tape recorder/player subunit, the CONNECT control command which establishes a connection between a signal source and a Serial Bus plug is mandatory. c) If D-VHS can record a partial transport stream which is taken from a multi-programmed transport stream provided from a Serial Bus plug using DEMUX functionality of the internal Tuner subunit, the CONNECT control command which establishes a connection between a Serial Bus input plug and a demux destination plug of Tuner subunit is mandatory. d) Refer to Clause 3.8 for the internal configuration of D-VHS. e) When a supported CONNECT control command requesting a connection which is currently available is received, a response of ACCEPTED shall be returned. f) The *lock* bit set to one is not used.

Unit command	opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
Control command				
INPUT PLUG SIGNAL FORMAT	19h	O	M	a) Supporting INPUT PLUG SIGNAL FORMAT control command enables a controller to use INPUT PLUG SIGNAL FORMAT specific inquiry command to confirm whether a certain input signal format of the Serial bus plug (e.g. MPEG-2 TS, DVCR etc.) is implemented. b) If a specified signal format is different from the internal configured signal format, D-VHS shall refuse INPUT PLUG SIGNAL control command and return a response of REJECTED or NOT_IMPLEMENTED. In other words, the configuration shall not be changed by the INPUT PLUG SIGNAL FORMAT control command. c) D-VHS that can receive two or more kinds of signal format of a data stream, e.g. MPEG2-TS and DVCR, shall change input plug signal configuration according to the CIP header of the input data stream. d) If a specified signal format for receiving is identical with a designated signal format, a response of ACCEPTED shall be returned. e) If the internal signal format is configured to MPEG, and a same MPEG signal format is specified using the INPUT PLUG SIGNAL FORMAT control command, D-VHS shall accept this command regardless of a value of TimeShiftFlag .

Unit command	opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
Control command				
OUTPUT PLUG SIGNAL FORMAT	18h	O	M	a) Supporting OUTPUT PLUG SIGNAL FORMAT control command enables a controller to use OUTPUT PLUG SIGNAL FORMAT specific inquiry command to confirm whether a certain output signal format of the Serial bus plug (e.g. MPEG-2 TS, DVCR etc.) is implemented. b) This command is used to specify the signal format of the isochronous data stream, which is, transmitted to an output plug of the target (D-VHS). c) If D-VHS has the function that is able to convert a represented MPEG stream to a DV stream and is able to transmit the DV stream to an output plug instead of the MPEG stream, the signal format of the isochronous data stream to output is able to be specified by using this command. d) If a point-to-point connection(s) exists on the specified Serial Bus output plug, D-VHS should not change the output plug signal format by this command. **NOTE:** Since the definition for a bandwidth allocation in IEC61883 is not clear, the following clarification should be added. If a bandwidth reallocation is needed when D-VHS changes its signal format (or when the bit rate is increased), the D-VHS shall allocate the required bandwidth. e) If a specified signal format is the same as the internal configured signal format, D-VHS shall accept OUTPUT PLUG SIGNAL control command and return a response of ACCEPTED.

Unit command		opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
Control command					
OUTPUT PLUG SIGNAL FORMAT		18h	O	M	f) If the internal signal format is configured to MPEG, and a same MPEG signal format is specified using the OUTPUT PLUG SIGNAL FORMAT control command, D-VHS shall accept this command regardless of a value of TimeShiftFlag
POWER	power on	B2h	O	O	
	power off		O	M	
Status command					
CONNECT		24h	O	M	a) The CONNECT status command which inquires the source connected to a destination plug of a tape recorder/player subunit is mandatory. b) The CONNECT status command which inquires source connected to a Serial Bus output plug is mandatory. c) The CONNECT status command which inquires the source connected to an external output is mandatory. d) If a multi-programmed transport stream provided from a Serial Bus input plug can be de-multiplexed into a partial transport stream in a Tuner subunit, the CONNECT status command which inquires the source connected to a demux_destination_plug of Tuner subunit is mandatory. e) The CONNECT status commands except for the case of a) to d) are supported optionally. f) Refer to Clause 3.8 for the D-VHS internal connection configurations.

Unit command	opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
Status command				
INPUT PLUG SIGNAL FORMAT	19h	R	M	a) When D-VHS receives this command, D-VHS returns the response frame, which indicates the signal format that the Serial Bus input plug is configured to receive. As a signal format, which is used in the *fmt* field of the response frame, a value of 111111_2 (no data) is not used.
OUTPUT PLUG SIGNAL FORMAT	18h	R	M	a) When D-VHS receives this command, D-VHS returns the response frame, which indicates the signal format that the Serial Bus output plug is configured to transmit. As a signal format, which is used in the *fmt* field of the response frame, a value of 111111_2 (no data) is not used. b) If D-VHS is in playback mode and represents the MPEG stream from a tape medium, a TimeShiftFlag bit in the response frame of this command shall be set to "1". In case of other than playback mode, a TimeShiftFlag bit in the response frame of this command should be set to the appropriate value.
SUBUNIT INFO	31h	M	M	
UNIT INFO	30h	M	M	The parameter values of the response frame are as follows. The *unit_type* field value is 4h. The *unit* field value is 0h. The *company_ID* value is vendor-dependent.
POWER	B2h	O	M	

Unit command	opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
Notify command				
POWER	B2h	R	M	a) The command queue is level 1 or more. b) If D-VHS receives this command and the quantities of these commands, which are not executed, exceed the command queue level, D-VHS should return a response of REJECTED to the node, which has been issued the current executing command and returns a response of INTERIM to the node, which has been issued the latest command. **NOTE:** The bus traffic may be increased too much by re-sending this command to the same target right after a response of REJECTED is returned from that target. The definition for handling the notify command is to be clarified in the 1394TA.

Tape Recorder/Player Subunit Command		opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
CONTROL command					
BACKWARD	video frame	56h	O	O	
	video scene		O	O	
	VISS		O	O	
	GOP for MPEG recording		O	O	
	Index		O	O	
	Skip		O	O	
	Photo/Picture		--	--	
	Program_start		O	O	
	Random_marker		O	O	
FORWARD	video frame	55h	O	O	
	video scene		O	O	
	VISS		O	O	
	GOP for MPEG recording		O	O	
	Index		O	O	

<table>
<tr><th colspan="2">Tape Recorder/Player Subunit Command</th><th>opcode/ operand [0]</th><th>AV/C Support Level</th><th>D-VHS Support Level</th><th>Comment</th></tr>
<tr><td colspan="6">CONTROL command</td></tr>
<tr><td rowspan="4">FORWARD</td><td>Skip</td><td rowspan="4"></td><td>O</td><td>O</td><td></td></tr>
<tr><td>Photo/Picture</td><td>--</td><td>--</td><td></td></tr>
<tr><td>Program_start</td><td>O</td><td>O</td><td></td></tr>
<tr><td>Random_marker</td><td>O</td><td>O</td><td></td></tr>
<tr><td rowspan="2">RELATIVE TIME COUNTER</td><td>search</td><td>57h/20h</td><td>R</td><td>O</td><td></td></tr>
<tr><td>data</td><td>57h/71h</td><td>R</td><td>M</td><td>A value of Zero is the only mandatory value for resetting the counter.</td></tr>
<tr><td rowspan="9">INPUT SIGNAL MODE</td><td>D-VHS Digital</td><td>79h/01h</td><td>O</td><td>M</td><td rowspan="9">a) The INPUT SIGNAL MODE control command is used to specify the signal mode for recording of the Tape recorder/player subunit.
b) The command, which specifies the implemented recording mode, is mandatory.
NOTE: If a Tape recorder/player subunit does not support certain recording mode, the corresponding INPUT SIGNAL MODE control command is not mandatory.
c) Supporting INPUT SIGNAL MODE control command enables a controller to use INPUT SIGNAL MODE specific inquiry command to confirm whether a certain input signal mode of Tape recorder/player subunit is implemented.</td></tr>
<tr><td>VHS NTSC 525/60</td><td>79h/05h</td><td>O</td><td>M</td></tr>
<tr><td>VHS M-PAL 525/60</td><td>79h/25h</td><td>O</td><td>M</td></tr>
<tr><td>VHS PAL 625/50</td><td>79h/A5h</td><td>O</td><td>M</td></tr>
<tr><td>VHS N-PAL 625/50</td><td>79h/B5h</td><td>O</td><td>M</td></tr>
<tr><td>VHS SECAM 625/50</td><td>79h/C5h</td><td>O</td><td>M</td></tr>
<tr><td>VHS ME-SECAM 625/50</td><td>79h/D5h</td><td>O</td><td>M</td></tr>
<tr><td>S-VHS 525/60</td><td>79h/0Dh</td><td>O</td><td>M</td></tr>
<tr><td>S-VHS 625/50</td><td>79h/EDh</td><td>O</td><td>M</td></tr>
</table>

<table>
<tr><th colspan="2">Tape Recorder/Player Subunit Command</th><th>opcode/ operand [0]</th><th>AV/C Support Level</th><th>D-VHS Support Level</th><th>Comment</th></tr>
<tr><td colspan="6">CONTROL command</td></tr>
<tr><td rowspan="10">OUTPUT SIGNAL MODE</td><td>D-VHS Digital</td><td>78h/01h</td><td>O</td><td>M</td><td rowspan="10">a) The command, which specifies the implemented signal mode for playback, is mandatory.
b) Supporting OUTPUT SIGNAL MODE control command enables a controller to use OUTPUT SIGNAL MODE specific inquiry command to confirm whether a certain output signal mode of Tape recorder/player subunit is implemented.
c) If a specified signal mode for playback corresponds to an actual signal mode in playback, a response of ACCEPTED shall be returned; otherwise a response of REJECTED or NOT_IMPLEMENTED shall be returned. Therefore, the designated signal mode for playback shall not be changed by the OUTPUT SIGNAL MODE control command.
d) The contents (format) of the tape define the OUTPUT SIGNAL MODE.</td></tr>
<tr><td>VHS NTSC 525/60</td><td>78h/05h</td><td>O</td><td>M</td></tr>
<tr><td>VHS M-PAL 525/60</td><td>78h/25h</td><td>O</td><td>M</td></tr>
<tr><td>VHS PAL 625/50</td><td>78h/A5h</td><td>O</td><td>M</td></tr>
<tr><td>VHS N-PAL 625/50</td><td>78h/B5h</td><td>O</td><td>M</td></tr>
<tr><td>VHS SECAM 625/50</td><td>78h/C5h</td><td>O</td><td>M</td></tr>
<tr><td>VHS ME-SECAM 625/50</td><td>78h/D5h</td><td>O</td><td>M</td></tr>
<tr><td>S-VHS 525/60</td><td>78h/0Dh</td><td>O</td><td>M</td></tr>
<tr><td>S-VHS 625/50</td><td>78h/EDh</td><td>O</td><td>M</td></tr>
</table>

Tape Recorder/Player Subunit Command		opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
CONTROL command					
PLAY	NEXT FRAME	C3h/30h	R	O	
	SLOWEST FORWARD	C3h/31h	R	O	
	SLOW FORWARD 6	C3h/32h	O	O	
	SLOW FORWARD 5	C3h/33h	O	O	
	SLOW FORWARD 4	C3h/34h	O	O	
	SLOW FORWARD 3	C3h/35h	O	O	
	SLOW FORWARD 2	C3h/36h	O	O	
	SLOW FORWARD 1	C3h/37h	O	O	
	X1	C3h/38h	O	O	
	FAST FORWARD 1	C3h/39h	O	M	TP4/TP6/TP12
	FAST FORWARD 2	C3h/3Ah	O	M	TP4/TP6/TP12
	FAST FORWARD 3	C3h/3Bh	O	M	TP6/TP12
	FAST FORWARD 4	C3h/3Ch	O	M	TP6/TP12
	FAST FORWARD 5	C3h/3Dh	O	M	TP12
	FAST FORWARD 6	C3h/3Eh	O	M	TP12
	FASTEST FORWARD	C3h/3Fh	M	M	TP12/TP24
	PREVIOUS FRAME	C3h/40h	R	O	
	SLOWEST REVERSE	C3h/41h	R	O	
	SLOW REVERSE 6	C3h/42h	O	O	
	SLOW REVERSE 5	C3h/43h	O	O	
	SLOW REVERSE 4	C3h/44h	O	O	
	SLOW REVERSE 3	C3h/45h	O	O	
	SLOW REVERSE 2	C3h/46h	O	O	
	SLOW REVERSE 1	C3h/47h	O	O	

Tape Recorder/Player Subunit Command		opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
CONTROL command					
PLAY	X1 REVERSE	C3h/48h	O	O	
	FAST REVERSE 1	C3h/49h	O	M	TP4R/TP6R/TP12R
	FAST REVERSE 2	C3h/4A h	O	M	TP4R/TP6R/TP12R
	FAST REVERSE 3	C3h/4Bh	O	M	TP6R/TP12R
	FAST REVERSE 4	C3h/4Ch	O	M	TP6R/TP12R
	FAST REVERSE 5	C3h/4D h	O	M	TP12R
	FAST REVERSE 6	C3h/4Eh	O	M	TP12R
	FASTEST REVERSE	C3h/4Fh	M	M	TP12R/TP24R
	REVERSE	C3h/65h	O	O	
	REVERSE PAUSE	C3h/6D h	O	O	
	FORWARD	C3h/75h	M	M	Playback at normal speed
	FORWARD PAUSE	C3h/7D h	M	M	Pause in playback

The following table illustrates a relationship between trick play modes represented by TPxx or TPxxR and an actual trick play speed in each recording mode of D-VHS.

NOTE: X4 means that a forward playback at four times of normal speed.
X4R means that a reverse playback at four times of normal speed.

Trick play mode	Recording mode of D-VHS					
	STD mode	HS mode	LS2 mode	LS3 mode	LS5 mode	LS7 mode
TP4	X4	--	X8	X12	X20	X28
TP6	X6	X3	X12	X18	X30	X42
TP12	X12	X6	X24	X36	X60	X84
TP24	X24	X12	X48	X72	X120	X168
TP4R	X4 REVERSE	--	X8 REVERSE	X12 REVERSE	X20 REVERSE	X28 REVERSE
TP6R	X6 REVERSE	X3 REVERSE	X12 REVERSE	X18 REVERSE	X30 REVERSE	X42 REVERSE
TP12R	X12 REVERSE	X6 REVERSE	X24 REVERSE	X36 REVERSE	X60 REVERSE	X84 REVERSE
TP24R	X24 REVERSE	X12 REVERSE	X48 REVERSE	X72 REVERSE	X120 REVERSE	X168 REVERSE

The following two tables will help to clarify the various combinations of trick play mode and the PLAY control command:

PLAY control command	Trick play speed to be performed						
Playback_mode	Supported trick play mode (s) for D-VHS Digital FORWARD playback						
	Not Implementation	TP12	TP12 & TP4	TP12 & TP6	TP12 & TP24	TP12 & TP4 & TP24	TP12 & TP6 & TP24
FAST FORWARD1	An actual tape speed depends on an implemented speed. (See the NOTE below.)	TP12	TP4	TP6	TP12	TP4	TP6
FAST FORWARD2		TP12	TP4	TP6	TP12	TP4	TP6
FAST FORWARD3		TP12	TP12	TP6	TP12	TP12	TP6
FAST FORWARD4		TP12	TP12	TP6	TP12	TP12	TP6
FAST FORWARD5		TP12	TP 12	TP 12	TP12	TP12	TP12
FAST FORWARD6		TP12	TP 12	TP 12	TP12	TP12	TP12
FASTEST FORWARD		TP12	TP 12	TP 12	TP24	TP24	TP24

PLAY control command	Trick play speed to be performed						
Playback_mode	Supported trick play mode (s) for D-VHS Digital REVERSE playback						
	Not Implementation	TP12R	TP12R & TP4R	TP12R & TP6R	TP12R & P24R	TP12R & TP4R &TP24R	TP12R & TP6R & P24R
FAST REVERSE1	An actual tape speed depends on an implemented speed. (See the NOTE below.)	TP12R	TP4R	TP6R	TP12R	TP4R	TP6R
FAST REVERSE2		TP12R	TP4R	TP6R	TP12R	TP4R	TP6R
FAST REVERSE3		TP12R	TP 12R	TP6R	TP12R	TP12R	TP6R
FAST REVERSE4		TP12R	TP 12R	TP6R	TP12R	TP12R	TP6R
FAST REVERSE5		TP12R	TP 12R	TP12R	TP12R	TP12R	TP12R
FAST REVERSE6		TP12R	TP 12R	TP12R	TP12R	TP12R	TP12R
FASTEST REVERSE		TP12R	TP 12R	TP12R	TP24R	TP24R	TP24R

In the D-VHS Specifications, the implementation of trick play modes is optional. However if D-VHS implements the trick play mode, supporting both TP12 and TP12R in STD mode are mandatory.

How to execute the trick play mode on reception of various PLAY control command is defined in the table above. Note that the same PLAY control command may result in a different trick play mode depending on the supported trick play modes.

NOTE: When a trick play mode is not implemented, D-VHS cannot detect the recorded trick play data on a tape medium. Therefore, if D-VHS receives a PLAY control command with FAST FORWARD/REVERSE *n*, then the D-VHS does not need to reproduce data from a tape medium, however D-VHS should transport the tape medium to the corresponding direction at faster speed than the normal playback speed. If D-VHS supports no FAST *n* modes, a response of NOT_IMPLEMENTED shall be returned for that command.

Tape Recorder/Player Subunit Command		opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
CONTROL command					
RECORD	AREA 2+3 INSERT	C2h/31h	R	O	
	AREA 1 INSERT	C2h/32h	R	O	
	AREA 1+2+3 INSERT	C2h/33h	R	O	
	AREA 3 INSERT	C2h/34h	O	O	
	AREA 2 INSERT	C2h/36h	O	O	
	AREA 1+2 INSERT	C2h/37h	O	O	
	AREA 1+3 INSERT	C2h/38h	O	O	
	AREA 2+3 INSERT PAUSE	C2h/41h	R	O	
	AREA 1 INSERT PAUSE	C2h/42h	R	O	
	AREA 1+2+3 INSERT PAUSE	C2h/43h	R	O	
	AREA 3 INSERT PAUSE	C2h/44h	O	O	
	AREA 2 INSERT PAUSE	C2h/46h	O	O	
	AREA 1+2 INSERT PAUSE	C2h/47h	O	O	
	AREA 1+3 INSERT PAUSE	C2h/48h	O	O	
	RECORD	C2h/75h	M	M	If a Tape recorder/player subunit has record capabilities, the recording modes for the RECORD control command are mandatory.
	RECORD PAUSE	C2h/7Dh	M	M	
WIND	HIGH SPEED REWIND	C4h/45h	O	O	
	STOP	C4h/60h	M	M	
	REWIND	C4h/65h	M	M	
	FAST FORWARD	C4h/75h	M	M	
RECORDING SPEED	Standard Speed	DBh/6Fh	O	M	D-VHS/VHS SP mode If a Tape recorder/player subunit does not have record capabilities, this control command is not supported.
	Speed 34	DBh/22h	O	O	VHS VP (1/5) mode (only NTSC)
	Speed 33	DBh/21h	O	O	VHS EP mode
	Speed 32	DBh/20h	O	O	VHS LP mode
TAPE PLAYBACK FORMAT		D3h	O	O	Specify the D-VHS digital playback format

Tape Recorder/Player Subunit Command	opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
CONTROL command				
TAPE RECORDING FORMAT	D2h	O	M	a) This command is used to specify the recording format, i.e., STD, HS, LS2, LS3, LS5, and LS7, of the Tape recorder/player subunit. b) The command, which specifies the implemented recording format, is mandatory. **NOTE:** If a Tape recorder/player subunit does not support certain recording mode, the corresponding TAPE RECORDING FORMAT control command is not mandatory. c) If a Tape recorder/player subunit does not have specified recording capabilities, a response of NOT_IMPLEMENTED shall be returned. d) If a field whose *af[n]* bit is one contains unsupported value, D-VHS shall return NOT_IMPLEMENTED response. e) D-VHS should accept this command regardless of the input signal mode, i.e., D-VHS does not need to return a response of REJECTED even if the input signal mode is other than D-VHS-digital.
STATUS command				
RELATIVE TIME COUNTER	57h	R	M	If any displayable Relative Time Counter value exists, the value should be returned. Else the response of REJECTED should be returned.
SEARCH MODE	50h	R	O	In the case of VISS search, D-VHS shall return the number of remaining VISS markers to the target. (D-VHS calculates the remaining VISS marker numbers by subtracting the detected number of VISS markers from the originally specified number.)
INPUT SIGNAL MODE	79h	M	M	

Tape Recorder/Player Subunit Command	opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
STATUS command				
OUTPUT SIGNAL MODE	78h	M	M	a) If a medium is not loaded in the subunit, a REJECTED response should be returned. b) If a medium is loaded but the subunit cannot judge the recorded signal format, a REJECTED response should be returned. c) When the subunit reproduces non-recorded area, a REJECTED response should be returned. d) When the subunit reproduces the digital recorded area, "D-VHS digital" should be returned. e) When the subunit reproduces a tape other than digital playback mode, analog recording format will be returned.
MEDIUM INFO	DAh	R	M	
RECORDING SPEED	DBh	O	M	

<table>
<tr><th>Tape Recorder/Player Subunit Command</th><th>opcode/ operand [0]</th><th>AV/C Support Level</th><th>D-VHS Support Level</th><th>Comment</th></tr>
<tr><td colspan="5">STATUS command</td></tr>
<tr><td>TRANSPORT STATE</td><td>D0h</td><td>M</td><td>M</td><td>a) Inquire the status of the medium in the transport mechanism.
b) If D-VHS implements the trick play mode in digital signal playback, it shall return a response, which includes the transport_mode and transport_state shown below when this command is received in trick play mode.
• The transport_mode value: C3h
• The transport_state value: Select one from the values below corresponding to the current playback speed.
3Ah: x4 speed
3Ch: x6 speed
3Eh: x12 speed (if x24 speed is supported)
3Fh: x12 speed (if x24 speed is not supported)
3Fh: x24 speed (if x24 speed is supported)
4Ah: x4reverse speed
4Ch: x6reverse speed
4Eh: x12reverse speed (if x24reverse speed is supported)
4Fh: x12reverse speed (if x24reverse speed is not supported)
4Fh: x24reverse speed (if x24reverse speed is supported)</td></tr>
<tr><td>TAPE PLAYBACK FORMAT</td><td>D3h</td><td>M</td><td>M</td><td>a) Inquire the digital playback format status. A REJECTED response should not be used.
b) If a digital playback format is unspecified, no-information-response frame, i.e. medium_type field value is "00001b"(it indicates D-VHS) and the other fields are filled with zero, shall be returned.</td></tr>
</table>

Tape Recorder/Player Subunit Command	opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
STATUS command				
TAPE RECORDING FORMAT	D2h	M	M	a) Inquire the recording format status. A REJECTED response should not be used. b) If a recording format is unspecified, no-information-response frame, i.e. *medium_type* field value is "00001b"(it indicates D-VHS) and the other fields are filled with zero, shall be returned. c) D-VHS should return a STABLE response with the current digital recording format regardless of the input signal mode, i.e., D-VHS does not need to return no-information-response even if the input signal mode is other than D-VHS-digital.
NOTIFY command				
TRANSPORT STATE	D0h	O	M	a) The command queue is level 1 or more. b) If D-VHS receives this command and the quantities of these commands, which are not executed, exceed the command queue level, D-VHS should return a response of REJECTED to the node, which has been issued the current executing command and returns a response of INTERIM to the node, which has been issued the latest command. **NOTE:** The bus traffic may be increased too much by re-sending this command to the same target right after a response of REJECTED is returned from that target. The definition for handling the notify command is to be clarified in the 1394TA.

VENDOR-DEPENDENT Command (Defined as Unit Command)	opcode/ operand [0]	AV/C Support Level	D-VHS Support Level	Comment
STATUS command				
TIMER	00h	O	O	Inquire whether D-VHS is in a timer mode where some operation, i.e. key operation, AV/C command, may be refused so that a timer recording cannot be disturbed.
VERSION	00h	O	M	Inquire the version number of guideline supported.

3.7.2 VENDOR-DEPENDENT command

Two commands, TIMER command and VERSION command, are defined by JVC. These commands are based on VENDOR-DEPENDENT command defined in the AV/C specification. Both commands support only the command type of STATUS, and the legacy D-VHS does not support them.

3.7.2.1 TIMER command

	msb							lsb
opcode	VENDOR-DEPENDENT (00h)							
operand[0]	Company_ID (00 80 88h)(JVC)							
operand[1]								
operand[2]								
operand[3]	00h							
operand[4]	20h							
operand[5]	00h							
operand[6]	TIMER (A1h)							
operand[7]	FFh							
operand[8]	FFh							

Figure 3-1 TIMER status command format

The TIMER status command is defined as a Unit command and used to inquire the current timer mode status of D-VHS.
The format of TIMER status command is illustrated in Figure 3-1. The response format is illustrated in Figure 3-2 below.

	msb							lsb
opcode	VENDOR-DEPENDENT (00h)							
operand[0]	Company_ID (00 80 88h)(JVC)							
operand[1]								
operand[2]								
operand[3]	00h							
operand[4]	20h							
operand[5]	00h							
operand[6]	TIMER (A1h)							
operand[7]	1	0	0	0	0	0	0	don't care
operand[8]	1	0	0	0	0	0	0	timer mode

Figure 3-2 TIMER status response format

The *timer mode* bit indicates whether D-VHS is in a timer mode condition where D-VHS prevents some key operations and AV/C command from disturbing a timer recording. A value of zero indicates that D-VHS is not in the timer mode while a value of one indicates that D-VHS is in the timer mode.
The LSB of operand[7] does not be cared because some legacy D-VHS supports this command and defines this bit for vendor use. Usually a value of zero should be used as the LSB of operand[7].

3.7.2.2 VERSION command

VERSION status command is defined as a Unit command and used to inquire the version number of the D-VHS 1394 guideline implemented. The format of this command is illustrated in Figure 3-3 below.

	msb							lsb
opcode	VENDOR-DEPENDENT (00h)							
operand[0]	Company_ID (00 80 88h)(JVC)							
operand[1]								
operand[2]								
operand[3]	00h							
operand[4]	20h							
operand[5]	00h							
operand[6]	VERSION (A0h)							
operand[7]	FFh							
operand[8]	FFh							

Figure 3-3 VERSION status command format

The VERSION response format is illustrated in Figure 3-4 below.

	msb							lsb
opcode	VENDOR-DEPENDENT (00h)							
operand[0]	Company_ID (00 80 88h)(JVC)							
operand[1]								
operand[2]								
operand[3]	00h							
operand[4]	20h							
operand[5]	00h							
operand[6]	VERSION (A0h)							
operand[7]	version_number							
operand[8]	revision_number							

Figure 3-4 VERSION response format

The *version_number* field indicates the implemented version number of this guideline. The *version_number* field is encoded in binary coded decimal (BCD) format where the each nibble represents a decimal digit. The *revision_number* field indicates the revision number of the implemented guideline. The *revision_number* field is encoded in binary coded decimal (BCD) format where the each nibble represents a decimal digit. D-VHS comply with this guideline version 1.00 shall use a value of 01h for the *version_number* field and a value of 00h for the *revision_number* field.

3.8 Definition of D-VHS Model (for AV/C Command)

3.8.1 D-VHS model and its Internal Connections

As a basic function, D-VHS can record either an analog signal or a digital data stream on the medium in the suitable recording signal mode, i.e. an analog recording mode or a digital recording mode, that the signal can be recorded. Furthermore, D-VHS can record an analog signal in the digital recording mode by the MPEG encoder if it exists. In addition, D-VHS can record a digital signal in the analog recording mode by the MPEG decoder if it exists.
Therefore, both signal connections within a unit and a recording signal mode of a Tape recorder/player subunit in the unit shall be controlled by AV/C commands in a common way in order to guarantee an interoperability between D-VHS and a controller. Then D-VHS AV/C Model is defined and is illustrated in Figure 3-5 below.

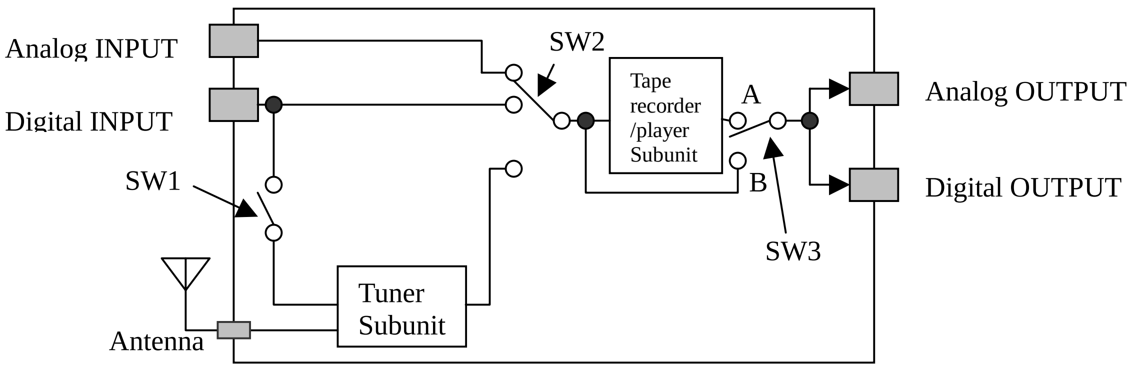

Figure 3-5 D-VHS AV/C Model

D-VHS AV/C Model consists of the followings.

a) Analog INPUT
b) Digital INPUT
c) Antenna (all kinds of antenna is included, e.g. terrestrial, satellite, etc)
d) Analog OUTPUT (an MPEG decoder may be included)
e) Digital OUTPUT (an MPEG encoder or a DV encoder may be included)
f) Tuner subunit (which has an antenna destination plug, a demux destination plug and one or more source plug for an analog signal and/or a digital data stream)

NOTE: To confirm the information for the Tuner subunit, i.e. analog tuner or digital tuner, a kind of antenna, etc, a controller should investigate the Tuner subunit identifier descriptor if it is implemented.

g) Tape recorder/player subunit (which has a destination plug and a source plug)
h) SW1 (connect or break between the Digital INPUT as an input plug of the unit and the demux destination plug of the Tuner subunit)
i) SW2 (select the signal source connected to the destination plug of the Tape recorder/player subunit)
j) SW3 (Usually SW3 state is changed according to the transport state of the Tape recorder/player subunit, i.e. select position A when D-VHS is in playback, select position B when D-VHS is in other than playback. If source monitor (EE) function is implemented,

D-VHS can select position A and B during playback by user key operation or CONNECT control command)

3.8.2 Selection of Signal source provided to a Tape recorder/player subunit

SW2 selects the signal source which can be connected to the destination plug of the Tape recorder/player subunit and a selection is specified by a CONNECT control command.
Possible signal sources are as follows.

a) Analog signal INPUT (this is selected by a CONNECT control command with a source plug defining *External input plug* of a Unit).

b) Digital signal INPUT (this is selected by a CONNECT control command with a source plug defining *Serial Bus iPCR* of a Unit).

c) Tuner Subunit (this is selected by a CONNECT control command with a source plug defining *Source plug0* of a Tuner subunit).

SW1 is disconnected when the tuner subunit does not make a partial stream from a received stream via Digital input.
The reason to designate SW1 is to simplify for getting the interconnection information by only the CONNECT status command.

3.8.3 Selection of recording signal mode for a Tape recorder/player subunit

INPUT SIGNAL MODE control command is used to select the recording signal mode of D-VHS, Analog signal recording mode such as VHS or S-VHS, Digital signal recording mode such as D-VHS Digital.

3.8.4 Selection of D-VHS output signal (Optional)

When D-VHS reproduces a signal from the tape, Both an analog signal OUTPUT and a Digital signal OUTPUT are usually connected to a source plug of a Tape recorder/player subunit without regard to the fact that, whether the Tape recorder/player subunit is in a digital playback or in an analog playback.

If D-VHS can output a source signal directly when a Tape recorder/player subunit is in the playback mode, SW3 may be changed from a Tape recorder/player subunit position to another position by accepting CONNECT control command. Otherwise, SW3 is changed automatically according to transport status of tape recorder/player subunit.

If CONNECT status command which inquires the source information of Unit output plug is received, the current output which is connected to the D-VHS output (one of 1) to 4) below) will be returned.

1) Source plug of a Tape recorder/player subunit
2) Analog input plug (Optional)
3) Digital input plug (Optional)
4) Source plug of tuner subunit (Optional)

3.9 Internal Connection Rule

3.9.1 CONNECT command and INPUT SIGNAL MODE command

3.9.1.1 Basic Policy:

Without regard to the fact that whether a cassette is inserted or not, keep a consistency between the recording mode (of a Tape recorder/player subunit) and an internal signal path (i.e. a signal path between any source plug in the unit and the destination plug of the Tape recorder/player subunit) so that any conflict can be avoided.

Example 1:

In the case of D-VHS which does not have an MPEG encoder, the recording signal mode of the Tape recorder/player subunit shall be an "analog" signal recording mode when the destination plug of the Tape recorder/player subunit is connected to the input plug which is configured to an analog signal (Analog INPUT).

Example 2:

If a VHS cassette is inserted in D-VHS which does not have an MPEG decoder, a CONNECT control command that specifies the digital signal input as the source plug and the Tape recorder/player subunit as the destination plug shall be refused and a response of REJECTED shall be returned.

3.9.1.2 Rule 1:

If a conflict between the internal signal path and the recording mode occurs due to having accepted a CONNECT control command, then the recording signal mode of the Tape recorder/player subunit shall be changed internally in order to avoid the conflict. (Note. in short, this means that a CONNECT control command overrides an INPUT SIGNAL MODE control command.)
However, if a cassette in Tape Recorder/Player Subunit cannot record the data specified by the CONNECT control command, then the command shall be refused and a response of REJECTED shall be returned.
In a case that there is no cassette inserted, if a CONNECT control command or an INPUT SIGNAL MODE control command which results in setting D-VHS into D-VHS-digital recording mode, is received, then the command may be refused and a response of REJECTED may be returned. However if the command is accepted and if a cassette inserted afterwards is capable to record only analog then a conflict between the recording mode and the capability of cassette may occur. If the conflict occurs, the solution is vendor dependent

If the requested signal path by the CONNECT control command has been already established, then a response of ACCEPTED shall be returned.

Example 3:

In the case of D-VHS which does not have an MPEG decoder, if the CONNECT control command which specifies the Digital INPUT as a signal source and the Tape recorder/player subunit as a destination plug is accepted, the recording signal mode shall be changed to the digital signal recording mode automatically. However, if a cassette whose medium is not capable for digital signal recording such as VHS is inserted, the

CONNECT control command shall be refused and a response of REJECTED shall be returned.

3.9.1.3 Rule 2:

If a conflict between the internal signal path and the recording mode occurs due to accepting an INPUT SIGNAL MODE control command, then the command shall be refused and a response of REJECTED shall be returned.

Example 4:

In the case of D-VHS which does not have an MPEG encoder for recording, if an INPUT SIGNAL MODE control command which requests a digital signal recording mode, is received when the analog INPUT is selected as a signal source, then this command shall be refused and a response of REJECTED shall be returned. (It shall be prohibited that the INPUT SIGNAL MODE control command is accepted and then the signal source is changed to the Digital INPUT internally)

To conform rules above, it becomes possible that specifying the signal source and recording signal mode of D-VHS is performed identically by the steps below.

Step0: (In case the cassette type which is inserted is unknown) Identification of medium
Step 1: Specifying of the signal source to record by performing CONNECT control command
Step 2: (If a signal converter, e.g. DV-->MPEG, Analog-->MPEG, is implemented)
Specifying of the recording signal mode by performing an INPUT SIGNAL MODE control command
Step 3: (If no other operation is needed) Performing of RECORD control command

3.9.2 For the program selection capable D-VHS

A rule for D-VHS which can record a specified single program from the digital input stream where a destination plug of a Tuner subunit is connected to the source plug as a digital signal input (In the case that SW1 is ON, SW1 is shown in Figure 8)

3.9.2.1 Rule 3:

When D-VHS records an incoming digital data stream from a source plug of the unit without having any program selection by a Tuner subunit, then the source plug of the unit shall be directly connected to the destination plug of the Tape recorder/player subunit.
If program selection for recording to the incoming digital stream from a source plug of the unit takes place at Tuner subunit, then the source plug of the unit shall be connected to the destination plug of the Tuner subunit and the source plug of the Tuner subunit shall be connected to the destination plug of Tape recorder/player subunit.

3.10 D-VHS Configuration ROM format

The example of D-VHS Configuration ROM format is illustrated below.

Bus_info_block				
Offset	0-7	8-15	16-23	24-31
0400h	04h	crc_length	rom_crc_value	
0404h	'1'	'3'	'9'	'4'
0408h	irmc, cmc, isc, bmc, pmc, res.	cyc_clk_acc	max_rec, res., max_ROM	generation, res., link_spd
040Ch	node_vendor_id (company_id)			chip_id_hi
0410h	chip_id_lo			
Root_directory				
0414h	root_length		CRC	
0418h	03h	module_vendor_id (company_id)		
041Ch	81h	textual_descriptor offset ((la0 - 41Ch)/4)		
0420h	17h	Model ID		
0424h	81h	model_name_textual_descriptor offset ((la1 - 424h)/4)		
0428h	0Ch	node_capabilities		
042Ch	D1h	unit_directory offset ((da0 - 42Ch)/4)		
Unit_directory				
da0	unit_directory_length		CRC	
da0+4h	12h	unit_spec_id (00A02Dh)		
da0+8h	13h	unit_sw_version (010001h)		
da0+Ch	17h	Model ID		
da0+10h	81h	model_name_textual_descriptor offset ((la1 - da0 -10h)/4)		
textual_descriptor_leaf				
la0	length (03h or 04h or 05h or 06h)		CRC	
la0+4h	spec_type (00h)	specifier_id (000000h)		
la0+8h	language_id (00000000h)			
la0+Ch	Vendor Name - 1 (minimal ASCII code)			
:	:			
la0+18h	Vendor Name - 4 (minimal ASCII code)			
textual_descriptor_leaf				
la1	length (03h or 04h or 05h or 06h)		CRC	
la1+4h	spec_type (00h)	specifier_id (000000h)		
la1+8h	language_id (00000000h)			
la1+Ch	Model Name - 1 (minimal ASCII code)			
:	:			
la1+18h	Model Name - 4 (minimal ASCII code)			

Figure 3-6 Example of D-VHS Configuration ROM format

da0: Unit_directory entry offset address
la0: textual_descriptor_leaf entry offset address (for Vendor Name)
la1: text_descriptor_leaf entry offset address (for Model Name)

The D-VHS Configuration ROM implementations conforming to this guideline shall meet the performance criteria specified in the following documents.

(a) IEC 61883-1, Clause 5.3 [R3].
(b) IEEE Std 1394a-2000, Clause 10.25 and 10.26 [R2].
(c) 1394 Trade Association, Configuration ROM for AV/C Devices 1.0 [R5].

Figure 3-6 shows the example of D-VHS Configuration ROM format. D-VHS Configuration ROM shall include the Bus_info_block, the Root_directory, the Unit_directory, and the textual descriptors in Minimal ASCII that describe the 'Vendor Name' and the 'Model Name'. The Root directory may include more entries if the optional descriptors for module_vendor_id (Vendor_ID) and/or Model_ID are implemented. Shaded addresses are newly defined in Configuration ROM for AV/C Devices 1.0 [R5] but not defined in IEC61883-1 [R3].

3.10.1 Vendor Name

The Vendor Name shall be described by the textual descriptor in Minimal ASCII. The Vendor Name length should be within 16 characters (bytes) and stuffing "00h" for remain area.
When only the Minimal ASCII descriptor is present, the leaf offset entry for the descriptor shall immediately follow the module_vendor_id (Vendor_ID) in the Root directory.
When more than one textual descriptor is present, the textual descriptor directory offset entry shall immediately follow the module_vendor_id (Vendor_ID) in the Root directory and the Minimal ASCII textual descriptor offset shall be placed in the first entry of the descriptor directory.

Example. Case of JVC

1) Model for countries except JAPAN JVC (uppercase letter)

In the case of eight characters (4Ah 56h 43h 00h 00h 00h 00h 00h)

2) Model for JAPAN VICTOR (uppercase letter)

In the case of eight characters (56h 49h 43h 54h 4Fh 52h 00h 00h)

3.10.2 Model ID

The Model_ID is the vendor-dependent data field to describe the model dependent information and is placed in the Root directory. The Model_ID shall immediately follow either, Textual descriptor leaf offset entry for module_vendor_id or, Textual descriptor directory offset entry for module_vendor_id or, ICON descriptor leaf offset entry for module_vendor_id.

3.10.3 Model Name

The Model Name shall be described by the textual descriptor in Minimal ASCII. The Model Name length should be within 16 characters (bytes) and stuffing "00h" for remain area.
When only the Minimal ASCII descriptor is present, the leaf offset entry for the descriptor shall immediately follow the Model_ID in the Root directory.
When more than one textual descriptors is present, the textual descriptor directory offset entry shall immediately follow the Model_ID in the Root directory, and the Minimal ASCII textual descriptor offset shall be placed in the first entry of the descriptor directory.

The Model_ID and its textual descriptor leaf offset entry are also added after the unit_sw_version entry in the unit directory because those would be used for the driver identification, which is defined in Microsoft Plug and Play Design Specification for IEEE 1394 Version 1.0c [R8].

Example. Case of JVC

1) Model Name HM-DR1 (uppercase letter)

In the case of sixteen characters

(48h 4Dh 2Dh 44h 52h 31h 00h 00h 00h 00h 00h 00h 00h 00h 00h 00h)

Annex.A. **Consideration for the legacy models**

a) Since there are D-VHS models, which are not compliant to this guideline, introduced in the market (hereinafter these models are called Legacy D-VHS). Therefore, this clause is important to take it into consideration that interchangeability with those legacy models is secured, too.

b) A functionally different point with the legacy model and the standardized model by this guideline is shown in the following.

c) The legacy model may not correspond to the D-VHS Configuration ROM format, e.g. no Vendor Name, no Model ID and no Model Name.

d) The legacy model may not implement the function, i.e. selecting a source node, establishing a point-to-point connection, as described in Clause 3.5 a).

e) When a D-VHS detects that there are only two nodes on the Serial Bus and the other node is also a D-VHS, and when the D-VHS is going into a Digital Playback mode, it should establish or overlay a point-to-point connection between its own oPCR and the iPCR of the other D-VHS.

NOTE1: The purpose of the e) is to guarantee that the legacy model can record an isochronous data stream providing from the standardized model on the Serial Bus establishing a point-to-point connection between the legacy model's iPCR and the standardized model's oPCR. Because the channel number, which is used in a broadcast INPUT/OUTPUT, may be different from each other in legacy models and the legacy models do not have the functionality described in section 3.5, those models may not be connected with the standardized models without a point-to-point connection. The standardized model may establish a point-to-point connection between its own oPCR and another standardized model's iPCR according to rule e).

NOTE2: When D-VHS is in the condition except a Digital Playback mode, establishing or overlaying a point-to-point connection may not be executed even if D-VHS can output an isochronous data stream by the internal STB (Set Top Box).
Digital Playback mode means the conditions that 1) D-VHS reproduces a digital signal recorded on the D-VHS medium, 2) D-VHS reproduces an analog signal recorded on the VHS/S-VHS/D-VHS medium and the internal MPEG encoder encodes that an analog signal to an MPEG stream and output it as a playback signal.

f) The legacy model may not implement the AV/C command, which is newly defined as a mandatory command for D-VHS. (e.g. CONNECT control/status command, TRANSPORT STATE/POWER notify, VERSION status command, etc)

g) The value of *fmt* field of the output signal may be set to "111111b", no data if the tape playback is in analog or no signal part.

Annex.B. **The way of recognition for D-VHS**

a) A controller is able to detect D-VHS on the Serial Bus by using a VERSION status command. If a response of STABLE is returned for the VERSION status command then the device is identified as a D-VHS.

b) If the target device is a legacy D-VHS that does not support this guideline, a VERSION status command is refused and a response of NOT_IMPLEMENTED can be returned. Then it can be recognized whether or not the target device is D-VHS by the following means.

c) Both TAPE RECORDING FORMAT status command and TAPE PLAYBACK FORMAT status command may be used to identify devices on the bus as D-VHS or not. If a device returns a response with other than NOT_IMPLEMENTED, then the device may be identified as D-VHS.

d) If both means described above were not succeeded, the target device cannot be D-VHS.